Jungang Zou

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Research Interests: Bayesian methodology; data integration and generalizability; statistical learning with incomplete and distributed data; AI-assisted statistical computing.

Education

2022–27*	**Ph.D.**, Biostatistics, Columbia University
2019-22†	**M.S.**, Biostatistics, Columbia University
2015-19	**B.E.**, Software Engineering, Xiamen University, China

Research Appointments

2021–	Graduate Research Assistant, Columbia University, (Advisor: Dr. Qixuan Chen)
2020–	Graduate Research Assistant, Rutgers University, (Advisor: Dr. Liangyuan Hu)
2018	Undergraduate Research Assistant, Harvard Medical School (Advisors: Dr. Fan Lin and Dr. Gil Alterovitz)

Awards and Honors

2026	Paper Award, STAI-X 2026 (one of 9 awards selected from all 54 accepted papers).
2026	Student Paper Competition Award, GSS/SRMS/SSS Sections, JSM 2026.
2026	Outstanding Doctoral Teaching Assistant in Biostatistics, Columbia University Mailman School of Public Health.
2026	Travel Grant, IMSI Workshop on New Horizons on Model Transportability and Data Integration.
2025	Travel Grant, 14th International Conference on Bayesian Nonparametrics.

*Expected.

†Gap year between 2020-21 due to the spread of Covid-19.

Publications

Journal Articles

J1. Flores, N. M., Casey, J. A., Lovinsky-Desir, S., Spira-Cohen, A., Louis, S., Walters, S., Little, M., **Zou, J.**, Rudolph, K. E., Hernández, D., Chen, Q. & Perzanowski, M. A18-10 Large-Scale Mold Intervention Reduced Asthma Emergency Department Visits in New York City Public Housing Differentially Across Building and Neighborhood Characteristics. *American Journal of Respiratory and Critical Care Medicine* **212,** aamag162.286 (May 2026).

J2. Flores, N., Casey, J. A., Lovinsky-Desir, S., Spira-Cohen, A., Luis, S., Walters, S., Little, M., **Zou, J.**, Rudolph, K., Hernández, D., Chen, Q. & Perzanowski, M. The Impact of a 2019 System-Wide Mold Intervention in NYC Public Housing on Emergency Department Visits for Asthma. *Journal of Allergy and Clinical Immunology* **157,** AB295 (Feb. 2026).

J3. Kim, K., **Zou, J.**, Feng, B., Luan, M., Jones, A., Gadbois, D., Lim, J., Schwartz, J. E., Chen, Q. & Moise, N. Predictors of Sustainability in the Collaborative Care Medicaid Program for Depression: A Cross-Sectional Study. *Psychiatric Services*, appi.ps.20250212 (2026).

J4. Zhang, K., Li, Y., **Zou, J.**, Wang, S., Fausto, B. A. & Hu, L. Multi-Level Variable Selection Using a BART-Enhanced Mixed-Effects Framework. *Statistics in Medicine* **45,** e70593 (2026).

J5. Flores, N. M., Lovinsky-Desir, S., Divjan, A., Hoepner, L. A., **Zou, J.**, Miller, R. L., Herbstman, J. B., Perera, F. P., Perzanowski, M. S. & Chen, Q. Trajectory analysis of rhinitis in a birth cohort from lower-income New York City neighborhoods. *Journal of Allergy and Clinical Immunology* **154,** 111–119 (2024).

J6. Greenleaf, A. R., Francis, S., **Zou, J.**, Farley, S. M., Lekhela, T., Asiimwe, F. & Chen, Q. Influenza-Like Illness in Lesotho From July 2020 to July 2021: Population-Based Participatory Surveillance Results. *JMIR Public Health Surveill* **10,** e55208 (2024).

J7. Onyilofor, C., Feng, B., **Zou, J.**, Barbecho, J. M., Duran, A. T., Ye, S., Garcia, O., Singer, J., Chen, Q. & Moise, N. Assessing the Fidelity of Depression Screening Implementation in Coronary Heart Disease Patients: A Cross-Sectional Study. *Journal of the American Heart Association* **13,** e035550 (2024).

J8. Williams, S. Z., **Zou, J.**, Liu, Y., Si, Y., Galea, S. & Chen, Q. Improving Survey Inference Using Administrative Records Without Releasing Individual-Level Continuous Data. *Statistics in Medicine* **43,** 5803–5813 (2024).

J9. Xu, Z., Scharp, D., Hobensack, M., Ye, J., **Zou, J.**, Ding, S., Shang, J. & Topaz, M. Machine learning-based infection diagnostic and prognostic models in post-acute care settings: a systematic review. *Journal of the American Medical Informatics Association* **32,** 241–252 (2024).

J10. Jadow, B. M., Hu, L., **Zou, J.**, Labovitz, D., Ibeh, C., Ovbiagele, B. & Esenwa, C. Historical Redlining, Social Determinants of Health, and Stroke Prevalence in Communities in New York City. *JAMA Network Open* **6,** e235875–e235875 (2023).

J11. Hu, L., **Zou, J.**, Gu, C., Ji, J., Lopez, M. & Kale, M. A flexible sensitivity analysis approach for unmeasured confounding with multiple treatments and a binary outcome with application to SEER-Medicare lung cancer data. *The Annals of Applied Statistics* **16,** 1014–1037 (2022).

Peer-reviewed Conference Proceedings

C1. Moise, N., Sales, A., Luan, M., Feng, B., Jones, A., Gadbois, D., **Zou, J.**, Kim, K., Schwartz, J. & Chen, Q. *Examining determinants of long-term reach and effectiveness following collaborative care implementation to optimize the pace of sustainability efforts.* in *Proceedings of the 17th Annual Conference on the Science of Dissemination and Implementation* (AcademyHealth, Arlington, VA, 2024).

Preprints and Manuscripts

P1. **Zou, J.** & Chen, Q. *Bayesian Federated Nonlinear Variable Selection under Site Heterogeneity with Unknown Sharing Structure.* Manuscript in preparation. 2026.

P2. **Zou, J.** & Hu, L. *Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies.* Under review @Computational Statistics and Data Analysis. 2026.

P3. **Zou, J.**, Jiang, A. Z. & Chen, Q. *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* 2026. arXiv: 2605.18476 [stat.CO].

P4. **Zou, J.**, Schwartz, J. E., Moise, N., Little, R. & Chen, Q. *Transporting Treatment Effects from Trials to EHR Populations using Propensity Score Predictive Inference.* Manuscript in preparation. 2026.

P5. **Zou, J.**, Wang, S. & Chen, Q. *Bayesian MI-LASSO for Variable Selection on Multiply-Imputed Data.* Under revision @Statistics in Medicine. 2026. arXiv: 2211.00114v2 [stat.ME].

P6. **Zou, J.**, Lin, F., Gao, S., Deng, G., Zeng, W. & Alterovitz, G. *Transfer Learning Based Multi-Objective Genetic Algorithm for Dynamic Community Detection.* 2021. arXiv: 2109.15136 [cs.SI].

Tools & Software

AI4BayesCode: From natural language to validated, modular, stateful Bayesian samplers. (Website | GitHub).

SBMTrees: Sequential Imputation with Bayesian Trees Mixed-Effects Models for Longitudinal Data (R | GitHub).

AuxSurvey: Survey Analysis with Auxiliary Discretized Variables (R | GitHub).

SAMTx: Sensitivity Assessment to Unmeasured Confounding with Multiple Treatments (R).

BMIselect: Bayesian MI-LASSO Models for Bayesian Variable Selection on Multiply-Imputed data (R | GitHub).

TMOGA: A Multi-Objective Genetic Algorithm for Dynamic Community Detection Problem (Pypi | GitHub).

Presentations

Invited Presentations

I1. **Zou, J.** *Bayesian Propensity Score Predictive Inference for Extending Trial Findings to Target Populations* SILVER Lab Meeting, University of Maryland School of Medicine. Forthcoming. Aug. 2026.

I2. **Zou, J.** *Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies* ENAR 2023 Spring Meeting (Nashville, TN, USA). Invited session: *Bayesian Machine Learning for Decision-Making with Incomplete Information*. Mar. 2023.

Contributed Talks

T1. **Zou, J.** *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* STAI-X 2026 (Boston, MA, USA). Aug. 2026.

T2. **Zou, J.** *Bayesian MI-LASSO for Variable Selection on Multiply-Imputed Data.* Joint Statistical Meetings 2026 (Boston, MA, USA). Aug. 2026.

T3. **Zou, J.** *Generalizing Treatment Effects from Trials to EHR Populations using Propensity Score Predictive Inference.* ENAR 2026 Spring Meeting (Indianapolis, IN, USA). Mar. 2026.

Posters

P1. **Zou, J.**, Jiang, A. Z. & Chen, Q. *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* STAI-X 2026 (Boston, MA, USA). Aug. 2026.

P2. **Zou, J.**, Schwartz, J. E., Little, R., Moise, N. & Chen, Q. *Transporting Treatment Effects from Randomized Trials to EHR Populations using Bayesian Propensity Score Predictive Inference.* IMSI Workshop: New Horizons on Model Transportability and Data Integration (Chicago, IL, USA). 2026.

P3. **Zou, J.** & Hu, L. *Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies.* 14th International Conference on Bayesian Nonparametrics (Los Angeles, CA, USA). June 2025.

P4. **Zou, J.**, Schwartz, J. E., Little, R., Moise, N. & Chen, Q. *Generalizing Treatment Effect from EHR-Recruited Trials Using Bayesian Propensity Prediction.* 2025 Columbia Biostatistics Annual Research Symposium (New York, NY, USA). Apr. 2025.

P5. **Zou, J.**, Wang, S. & Chen, Q. *Variable Selection for Multiply-imputed Data: A Bayesian Framework.* 2023 Columbia Biostatistics Annual Research Symposium (New York, NY, USA). Sept. 2023.

Teaching

Teaching Assistant, Columbia University

Spring 2025, 2026	Data Science II (P8106)
Fall 2021, 2023, 2025	Analysis of Health Surveys (P8123)
Fall 2024, 2026	Clinical Trial Methodology (P8142)

Summer 2023, 2024	Statistical Analysis with Missing Data Workshop
Spring 2024	Topics in Advanced Statistical Computing (P8160)
Spring 2023	Statistical Inference (P8109)
Fall 2022	Applied Regression II (P8110)

Academic Service

2026 Session Chair, "Sampling, Weighting, Missing Data and Variance Estimation," JSM 2026.

2026 Reviewer, *Computational Statistics*.

Other Experience

2020 Software Engineer Intern, WorldQuant, Beijing, China.

Last updated: August 12, 2026